Homework 2 – k Sample Inference

1 Question 1: Why do we Correct for Multiple Comparisons?

A student asked me what the point of correcting for multiple comparisons was. They made a two interesting points:

1. They are different outcomes, so, unlike the coin flip example (where the coin is the same) it doesn't makes sense to combine them.
2. If a manuscript presented three studies with t-tests, they don't correct for multiple comparisons across *those* tests.

When we conduct all pairwise comparisons after an omnibus test like ANOVA, we are testing a single family of hypotheses to answer one overarching question ("Which of these groups are different?"). Every additional test provides another opportunity to commit a Type I error from random noise. Similar to the coin, the probability of making at least one false error across J independent tests grows exponentially: $1-(1-\alpha)^J$. Note that our pairwise comparisons are not independent, so it's slightly different.

We don't correct for inferences across separate studies because they typically test distinct hypotheses using independent samples. Our attempt to address the multiple comparisons problem is specifically confined to the issue of increased Type I error within a single decision-making context. If we don't, we can find ourselves in a p-hacking-like situation, where an investigator might search through many tests to find one $p < 0.05$ (a potential false discovery), rather than making decisions to best position themselves for finding a true discovery.

Here, we will quickly demonstrate why we correct for the multiple comparisons problem when conducting several tests to answer the same question.

1.1 Part a

Complete the following steps in $R = 1000$ simulations.

1. Simulate data for $k = 5$ samples $n = 30$ drawn from a standard Gaussian distribution.
2. Use `pairwise.t.test(..., p.adjust.method="none")` to compute unadjusted pairwise t-tests and record if any comparison yields statistically discernible support for a non-zero mean.
3. Use `rstatix::tukey_hsd(...)` to compute the Tukey's HSD adjusted pairwise tests and record if any comparison yields statistically discernible support for a non-zero mean.

In the end, describe the Type I error rates.

```
n = 30
k = 5
R = 1:1000


significance.dat = tibble(simulation = R, unadjusted = rep(NA, 1000), adjusted = rep(NA, 1000))

for(i in R){
  dat = tibble(
```

```
    value = rnorm(n * k),
    group = as.factor(rep(1:k, each = n))
  )
  unadjusted = pairwise.t.test(dat$value, dat$group, p.adjust.method = 'none')
  adjusted = rstatix::tukey_hsd(dat, value ~ group)

  significance.dat$unadjusted[i] = any(unadjusted$p.value < 0.05, na.rm = TRUE)
  significance.dat$adjusted[i] = any(adjusted$p.adj < 0.05)
}

t1.unadj = length(which(significance.dat$unadjusted == TRUE))/length(R)
t1.adj = length(which(significance.dat$adjusted == TRUE))/length(R)
```

This simulation resulted in a type 1 error of 28.7% for the unadjusted t-tests, and a type 1 error. of 5% for the adjusted t-tests. This makes sense because without adjusting for Family Wise Error Rate (FWER) we expect the error rate to be $\alpha \times k$ ($0.05 \times 5 = 0.30$), which is exactly what we see in the simulation. When adjusting we expect the FWER to be α, and in this case the adjusted rate is approximately $\alpha = 0.05$.

1.2 Part b

Redo the simulation, changing to adjust according to the false discovery rate `p.adjust.method="BH"` and comment on the differences.

```
n = 30
k = 5

BH = rep(NA, 1000)

R = 1:1000
for(i in R){
  dat = tibble(
    value = rnorm(n * k),
    group = as.factor(rep(1:k, each = n))
  )
  bhadjust = pairwise.t.test(dat$value, dat$group, p.adjust.method = 'BH')

  BH[i] = any(bhadjust$p.value < 0.05, na.rm = TRUE)
}

t1.BH = length(which(BH == TRUE))/length(R)
```

Running the simulation again while using the 'BH' adjust method yields a type 1 error of 3.7% which is what we would expect for the adjusted FWER.

2 Question 2: Power Analyses

Revisit the MFAP4 study from class.

2.1 Part a

Assume the researchers were targeting a moderate effect size of Cohen's $f = 0.25$, setting significance level $\alpha = 0.05$ and desired power $1 - \beta = 0.80$. Were they adequately powered?

Note: The actual effect size is $f = 0.55$, but researchers usually do not know that ahead of time and defaulted to moderate, instead of specifying a large effect.

What is the required sample size per group (n) and the overall total sample size (N)? Write the `R` code using the `pwr` package to calculate this sample size. This package assumes all samples are of the same size so $N = nk$, where k is the number of samples. Note that the best answers will have a corresponding graphical display.

```r
library(pwr)
k = 5
f = 0.25
alpha = 0.05
power = 0.8

anova.power = pwr::pwr.anova.test(k = k, f = f, sig.level = alpha, power = power)

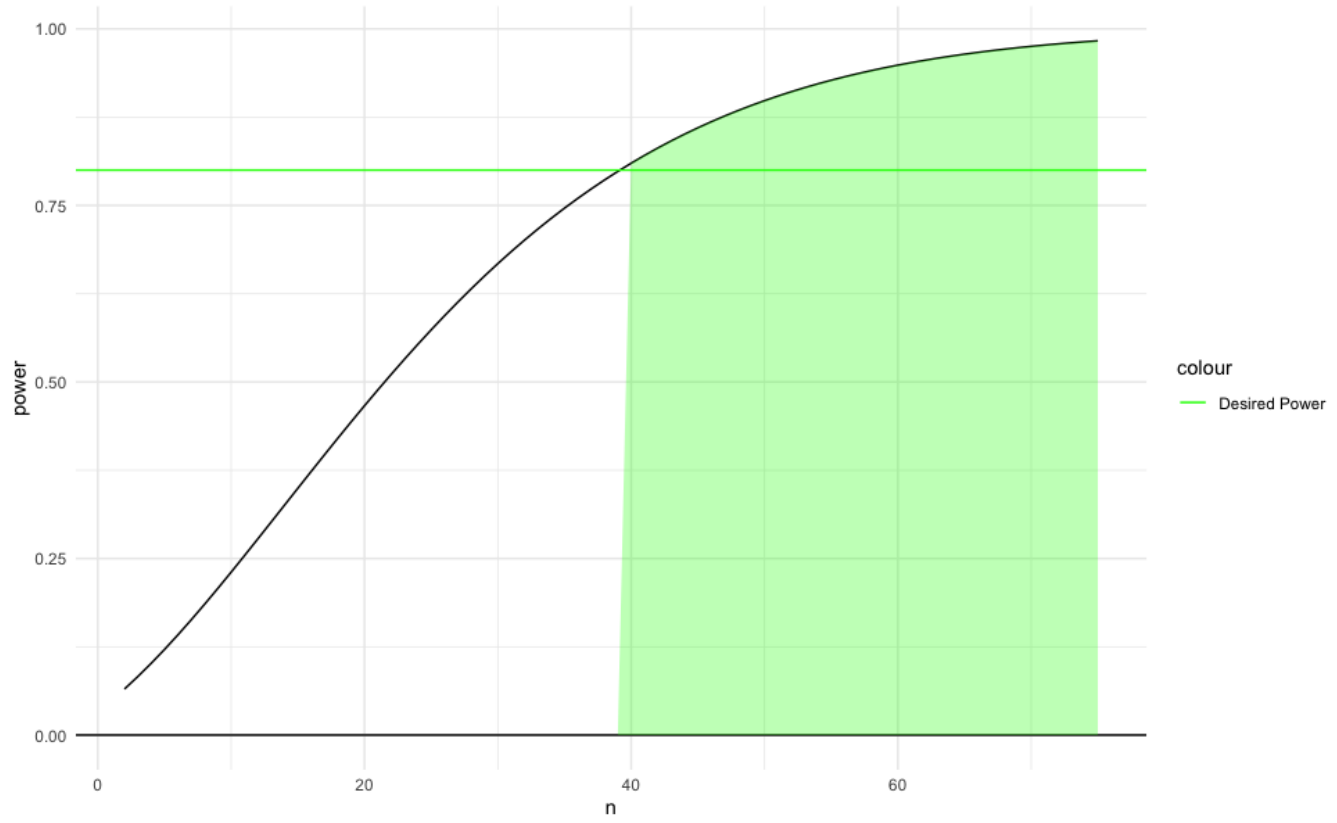
n = ceiling(anova.power$n)
N = n*k

power.dat = tibble(n = 2:75, power = rep(0, 74))

for(i in 2:75){
  power.dat$power[i-1] = pwr::pwr.anova.test(k = k, n = i, f = f, sig.level = alpha)$power
}

ggplot(data = power.dat)+
  geom_line(aes(x = n, y = power))+
  geom_hline(aes(color = "Desired Power", yintercept = 0.8))+
  geom_hline(yintercept = 0)+
  geom_ribbon(aes(ymin = 0, ymax = ifelse(power>=0.8,power,0), x = n), fill = "green", alpha = 0.3)+
  scale_color_manual(values = c("Desired Power" = "green", "Desired n" = "blue"))+
  theme_minimal()
```

Figure 1: Plotting power vs group sample size



This graph shows the relationship between the group sample size (n) and power. As n increases power increases aswell. On the graph the calculated n required to get power of atleast 80% (marked in green) is shown by the green shaded region.

Using the `pwr` package I was able to run the simulation above showing what group sample sizes will achieve what power. Additionally I calculated the required group sample size ($n = 40$) to achieve a power of 0.8 and total sample size ($N = 200$) assuming each group has the same sample size.

2.2 Part b

The project was funded by several agencies. Suppose that a funding agency suggested funding limiting the researchers to $N = 150$. Evaluate this suggestion in the context of the ANOVA. Would that be large enough to be adequately powered?

Compute the non-centrality parameter

$$\lambda = k \cdot n \cdot f^2$$

and use the non-central F distribution to compute the power under H_a for Cohen's $f = 0.25$. Calculate the statistical power achieved under this budget constraint. Note that the best answers will have a corresponding graphical display.

```
N = 150
k = 5
n = N/k
f = 0.25

lambda = N*(f^2)

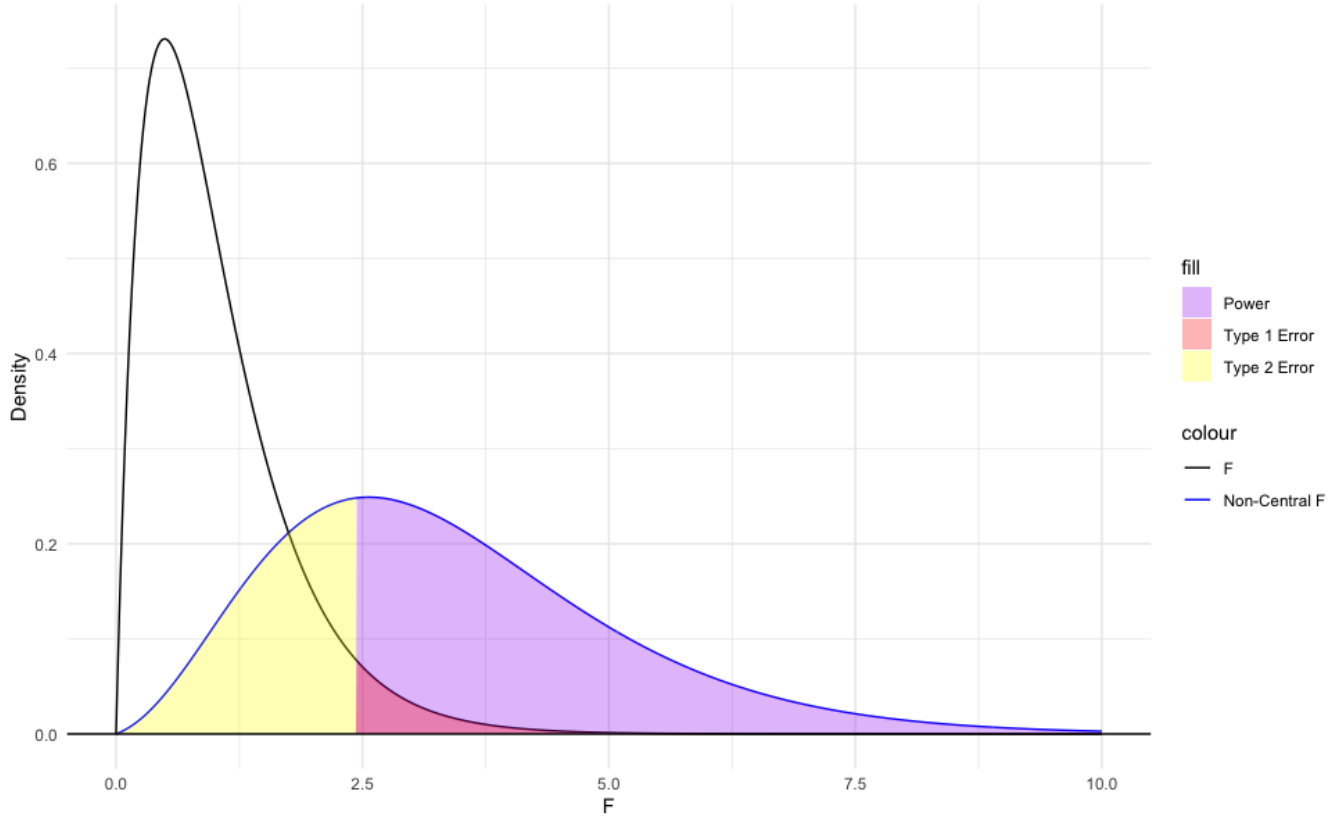
F = qf(0.95, k-1, N-k)
```

```r
power = pf(F, k-1, N-k, ncp = lambda, lower.tail = FALSE)

f.dat = tibble(f = seq(0, 10, length.out = 1000)) |>
  mutate(density1 = df(f, k-1, N-k, lambda),
         density2 = df(f, k-1, N-k))

ggplot(data = f.dat)+
  geom_line(aes(x = f, y = density1, color = "Non-Central F"))+
  geom_line(aes(x = f, y = density2, color = "F"))+
  geom_ribbon(aes(ymin = 0, ymax = ifelse(f > F, density1, 0), x = f, fill = "Power"), alpha = 0.3)+
  geom_ribbon(aes(ymin = 0, ymax = ifelse(f < F, density1, 0), x = f, fill = "Type 2 Error"), alpha = 0.3)+
  geom_ribbon(aes(ymin = 0, ymax = ifelse(f > F, density2, 0), x = f, fill = "Type 1 Error"), alpha = 0.3)+
  geom_hline(yintercept = 0)+
  labs(y = "Density", x = "F")+
  scale_color_manual(values = c("Non-Central F" = "blue", "F" = "black"))+
  scale_fill_manual(values = c("Power" = "purple", "Type 1 Error" = "red", "Type 2 Error" = "yellow"))+
  theme_minimal()
```

Figure 2: Visualizing Power using F-distributions



This graph shows the central F-distribution with parameters $df_1 = 4$ and $df_2 = 145$ and the non central F-distribution with the same degrees of freedom and non centrality parameter $\lambda = 9.38$, calculated for an effect size $f = 0.25$. Using the critical F calculated in the code above ($F_{critical} = 2.43$), the graph displays the power, and the type 1 and 2 errors if the total sample size were $N = 150$.

By computing the non centrality parameter $\lambda = 9.38$, and the critical F statistic $F_{critical} = 2.43$, I was able to compare the central F-distribution with the noncentral F-distribution which we assume is the population distribution. With $\alpha = 0.05$ and effect size $f = 0.25$, the expected power for a sample size $N = 150$ is 0.67. This is not sufficient because it is less than the desired power 0.8

2.3 Part c

In the article the researchers state:

> *However, the statistical power was not sufficient to allow an accurate discrimination of no to moderate stages of hepatic fibrosis (F0–F2) from severe fibrosis and cirrhosis (F3 and F4).*

The European Association for the Study of the Liver (EASL) provides treatment recommendations based on these stages.

- Stages 0-1: Treatment is deferred with regularly assessment
- Stage 2: Treatment is justified for those in stage 2
- Stage 3-4: Treatment is a priority for those in stages

Suppose the researcher wanted to perform a follow-up test where they want to ensure adequate power not just for the overall ANOVA F-test, but specifically to detect a pairwise differences between stages 1 and 2. Explain why the sample size computed in Part a may not be enough and use `PMCMRplus::power.tukey.test()` to compute a sample size that attains adequate power.

Note: The researchers would have had to do this *a priori*. Because we aren't medical experts, you can use sample estimates using the original data.

$$
\begin{aligned}
n &= \text{answer from part a} \\
\mu_\Delta &= 0.25 \\
\sigma^2_{\text{pooled}} &= \frac{(176-1)0.85^2 + (135-1)0.84^2}{176+135-2} = 0.7152 \\
\sigma_{\text{within}} &= 0.77
\end{aligned}
$$

2.4 Part d

How does the answer to Part c change if they wanted to have adequate power for *all* pairwise comparisons? Use `Superpower::ANOVA_design()` to set up the ANOVA set up.

Note: Use design "5b" and the following.

$$
\begin{aligned}
\mu &= \{2.92, 3.11, 3.36, 4.04, 4.35\} \\
\sigma^2 &= \frac{(97-1)0.87^2 + (176-1)0.85^2 + (135-1)0.84^2 + (67-1)1.00^2 + (67-1)0.90^2}{97+176+135+67+67-5} = 0.80
\end{aligned}
$$

Compute the power of the researcher's design (part a) using `Superpower::ANOVA_power()`. If the researchers were able to oversample by double, a common approach to addressing this issue, would their power be at least 80% for all pariwise comparisons?

Note: This code can take a minute. You may want to set `cache: true` in your `R` chunk yaml header. You will also see more power because the actual effect size was 0.55.

3 Question 3: An Application

Yang et al. (2023) investigate whether low-dose nicotine can counteract age-related metabolic decline. The proposed mechanism is through NAD^+, which is a chemical compound essential for metabolic energy production. The idea is that low-dose nicotine could help improve and/or maintain the recycling of NAD^+ by activating the salvage enzyme NAMPT via SIRT1. This could, in turn, restore NAD^+ homeostasis, which is the biological balance maintained within a cell between the production, recycling, and degradation of nicotinamide adenine dinucleotide (NAD^+).

Note: It is low-dose nicotine. Cigarette smoking delivers massive doses of nicotine along with other harmful chemicals. This is not an endorsement for smoking.

Male mice across distinct life stages – 2, 6, and 16 months – were categorized to evaluate natural, age-dependent metabolic and enzymatic shifts in two conditions.

- Treatment: low-dose nicotine (administered in drinking water)
- Control: no-dose placebo (administered in drinking water)

The researchers then aimed to compare treatment against control in the following samples:

- 2 months old, control
- 6 months old, control
- 16 months old, control
- 16 months old, treatment

Using PET imaging, the researchers measured age-induced inflammatory glucose hypermetabolism (SUV_{max}). They want to show that this quantity increases with age and those in the nicotine group saw levels comparable to younger mice.

Table 1: SUV_{max} measurements in the heart across age and treatment groups

Replicate	2M	6M	16M	16M, Treatment
1	1.48969	4.36000	7.30182	4.69304
2	1.30168	4.28807	7.12416	2.51326
3	1.30620	3.64184	8.28764	2.38019

3.1 Part a

State the null (H_0) and alternative (H_a) hypotheses for the omnibus test.

3.2 Part b

Perform some exploratory data analysis to provide empirical information about the researchers question.

3.3 Part c

Perform and report the results of the omnibus test. Make sure to evaluate the conditions for using the method you choose.

3.4 Part d

Conduct post-hoc pairwise comparisons to determine if age-induced inflammatory glucose hypermetabolism increase with age and whether low-dose nicotine reduces age-induced inflammatory glucose hypermetabolism.

3.5 Part e

Write a short paragraph directly addressing the researcher's hypothesis using your work in (a)-(d).

References

Yang, Liang, Junfeng Shen, Chunhua Liu, et al. 2023. "Nicotine Rebalances NAD+ Homeostasis and Improves Aging-Related Symptoms in Male Mice by Enhancing NAMPT Activity." *Nature Communications* 14 (1): 900.